

KijaniKiosk Capstone — Scope Document

Track A: Infrastructure-First

Problem Statement

kk-payments currently deploys to a single environment (kijani-project on local Minikube) with no staging isolation. Engineers testing changes have no way to validate against production-like configuration without risking the live namespace, and there is no automated gate preventing an untested change from reaching production. The capstone adds a kijani-staging namespace, an automated staging deploy with smoke test, and a human approval gate before any production deploy, closing the gap Nia flagged in the Week 9 production readiness assessment.

What I Will Build

- kijani-staging namespace, provisioned by Terraform (kubernetes provider) and configured by Ansible (kubernetes.core.k8s), isolated from kijani-project.
- Jenkins pipeline: auto-deploy to staging on merge to main, smoke test against the staging Service, human approval gate (Thursday's input step pattern) before production deploy.
- kk-payments running in staging with environment-specific ConfigMap (staging DB_HOST distinct from production) using the same Deployment manifest shape as production.
- Week 10 serverless receipt chain (kk-receipts, kk-processor, kk-notifier) triggered by kk-payments, proving the async processing path required by both capstone tracks.
- AI governance log (docs/ai-governance-log.md) documenting real AI-assisted troubleshooting from this build, including a corrected governance-checklist finding.

What Is Out of Scope

- Prometheus alerting — a log-based error-rate alternative (Week 7 pattern) is allowed by the brief but not yet implemented in this repository as of submission; named explicitly as a Production Gap rather than silently omitted.
- Real cloud deployment — MinIO substitutes for S3 throughout, consistent with the course's documented no-cloud-account policy from Week 4 onward.

Success Criteria

- A push to main triggers an automated staging deployment with no manual steps other than the approval gate itself.
- The production approval gate requires an explicit named-user approval with a recorded APPROVAL_REASON, and blocks the production deploy stage until that approval is given — demonstrated live with screenshots of both the paused and completed states.
- The Week 10 serverless receipt chain fires end-to-end from a single trigger, with correlation (orderId/key) traceable through all three functions' structured logs.

Architecture Diagram

